Haeun Kim

Curriculum Vitae

Seoul, South Korea — hajuli@skku.edu

EDUCATION & EMPLOYMENT HISTORY

Bachelor of Systems Management Engineering (Expected 2029)
Sungkyunkwan University
Relevant Coursework: Calculus I, II; General Physics I; General Physics Laboratory I; Basis and Practice in Programming; Scientific & Technical Writing; Creative and Interdisciplinary Design
Current Coursework: Applied Statistics I: Theory and Practice; Operations Research I; Introduction to Management of Technology; Smart Factory Capstone Design I; Introduction to Data Analysis and AI; Discrete Mathematics
GPA: 4.32/4.5

Team Leader, Scientific Writing Project
Time Period: Nov 2025 – Dec 2025
Sungkyunkwan University
- Led a team project on research proposal development
- Managed project timeline and coordinated team meetings
- Developed a research proposal on microplastic adsorption using Spirulina–pectin complexes, including literature review and experimental design

External Relations Director, Volunteer Club
Time Period: Mar 2026 – Present
Sungkyunkwan University Rotaract Club
- Coordinate external partnerships for volunteer activities
- Participate in external meetings and organize collaborative events

Public Relations Member, Volunteer Club
Time Period: Sep 2025 – Feb 2026
Sungkyunkwan University Rotaract Club
- Designed social media promotional materials
- Operated on-site promotion booths for recruitment and events

Math Tutor (Part-time)
Time Period: Dec 2024 – Present
Private Academy
- Provided one-on-one math tutoring and academic support
- Managed student progress and study plans
- Assisted with administrative tasks and classroom management

Volunteer Math Mentor
Time Period: Sep 2025 – Nov 2025; Mar 2026 – Apr 2026
Suwon High School
- Conducted one-on-one mentoring sessions (2 hours per week)
- Supported students' understanding of core math concepts

PUBLICATIONS

Non-Reviewed Articles

- "A Study on the Legitimacy of the Art-in-Architecture Policy" (Academic Writing Coursework)
- "Cultural Transformation of the 'Shadow' Archetype: A Comparative Study of Nibelungenlied and Janghwa Hongryeon" (Understanding the Culture of German Course)
- "On Existence within Worthlessness" (Introduction to Philosophy Coursework)
- "Understanding Through Proper Distance: *Shoko's Smile* by Choi Eun-young" (Introduction to Literature Coursework)

Manuscripts in Preparation

- TomatoDiag: Multimodal DINOv2 and TimesNet Based 3-Integrated Diagnosis System for Smart Tomato Farming (planned submission to ACCV)

RESEARCH OUTPUTS

Patent

- Image and Sensor Multimodal Fusion-Based Smart Farm Tomato Integrated Diagnosis System (Patent application)

Software

- Tomato Fruit Maturity and Cultivation Environment Integrated Diagnosis Program (Registered software copyright)

CREATIVE ACTIVITY

Presentations

- Presented multiple academic and project-based works in class settings
- Led team presentations in literature and interdisciplinary courses

Projects

Smart Patrol Robot Prototype Project
- Developed a conceptual prototype addressing public safety issues
- Produced a promotional video as part of a team project

Experimental Study on Alcohol Absorption
- Designed and conducted an experiment comparing alcohol absorption (fasted vs. fed state)
- Analyzed results and wrote a scientific report

Literature Analysis Project: To Kill a Mockingbird
- Conducted in-depth analysis of minor characters in the novel
- Presented findings through a team-based presentation

TomatoDiag: Multimodal Multi-task Integrated Diagnosis Framework for Smart Tomato Farming
- Developed a multimodal AI framework integrating tomato images with EC and pH sensor data
- Built a unified model for tomato growth stage classification (Breaker, Pink, Red), nutrient solution prediction, and sensor anomaly detection
- Contributed to model architecture design, training pipeline implementation, and technical report writing

S-Global Challenger (University Extracurricular Program)
- Proposed a Korean integrated governance model for large-scale crowd management through a comparative analysis of the UK's Green Guide governance framework

Selected Research Literature

Team Literature Review
- Interpretable Deep Multimodal-Based Tomato Disease Diagnosis and Severity Estimation
- A Systematic Review on Deep Learning Techniques for Maturity Classification of Tomato Fruit on Plants

Individual Literature Review
- Multi-modal Deep Learning Improves Grain Yield Prediction
- Generative Semantic Communication: Diffusion Models Beyond Bit Recovery

SKILLS

Programming: Python (Basic), C (Basic)
Tools: Excel (Basic), Tracker (video analysis)
Other: PowerPoint, Basic Data Analysis

SCHOLARSHIPS & HONORS

- EBS Dream Scholarship (External), 3,000,000 KRW
- Academic Excellence Scholarship, Sungkyunkwan University

SERVICE & PROFESSIONAL DEVELOPMENT

Service to the Community and Public

- Active member of university volunteer club
- Participated in mentoring and community service activities